

MATH 13–Skeletal Notes

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Preface

These are **skeletal** notes. Every section carries the definitions, the statements of the theorems, and the outline of each example and each proof — but the details are left blank, and we fill them in together in class. What you are holding is the scaffolding of the course; the mathematics gets written into it as we do it.

So bring them with you. Before each class, print the section we are about to cover, or download it and annotate it on a tablet — the “PDF” button in the navigation bar gives you the whole book as one file to print from. Then complete the blanks in class as we work through the material. A page you filled in yourself is worth more later than any set of notes handed to you finished.

Everything for this course lives here: the notes themselves, the assignments, the review problems for each exam, and the solutions. Each item is added as we reach it, so check back regularly rather than assuming a page is final. Solutions to an assignment are posted only after its due date has passed; the review sets are posted the same way.

The problems are all gathered in the Exercises chapter. There are ten assignments, which are graded, and four sets of review problems, which are not — the review sets collect everything the assignments left over, and they are fair game on an exam. Assignments are due at 11:59 PM on the day shown below.

Assignments	Due day, at 11:59 PM
1, 2, 4, 5, 7, 8, 10	Friday
3, 6, 9	Wednesday (exam weeks 3, 6, and 9)

You hand your work in through Gradescope, not through this site. The steps — how to upload a scan or a photo of your written solutions, and how to tell Gradescope which page holds which problem — are under [Submitting Your Assignments in Gradescope](#) in the course syllabus. Read them once before the first assignment is due; a submission with the pages mismatched is the most common way points get lost for reasons that have nothing to do with the mathematics.

One more thing worth knowing before you start: each of the ten assignment pages carries a Socratic AI tutor that will coach you when you are stuck, without ever giving an answer away. How it works, what it will and will not do, and what to keep out of the chat are all set out under Using the AI Tutor, at the top of the Exercises chapter.

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Chapter 1

Hyperbolic Functions

Objectives

After completing this section, you should be able to:

- Define the hyperbolic sine and cosine in terms of e^x and e^{-x} , and differentiate them.
- Sketch the graphs of $\sinh x$ and $\cosh x$, and state their domains and ranges.
- Use the identity $\cosh^2 x - \sinh^2 x = 1$ and relate it to the parametrization of the hyperbola $x^2 - y^2 = 1$.
- Work with the remaining hyperbolic functions $\tanh x$, $\coth x$, $\operatorname{sech} x$, and $\operatorname{csch} x$, and apply hyperbolic identities to integrals and equations.

This chapter is about certain combinations of e^x and e^{-x} which appear in multiple branches of science and engineering. In many ways they are analogous to the trigonometric functions, and they have a similar relationship to the hyperbola that the trigonometric functions have to the circle, which is why they are known as **hyperbolic functions**. We will study these relationships gradually during this chapter.

These are skeletal notes: the definitions, identities, and problem statements are given, and blank boxes and empty grids are left wherever a derivation, a solution, or a sketch belongs, so that you can fill them in by hand.

1.1 Definitions and Derivatives

The **hyperbolic cosine** is defined as

$$\cosh x = \frac{e^x + e^{-x}}{2}, \quad (1.1.1)$$

and the **hyperbolic sine** is defined as

$$\sinh x = \frac{e^x - e^{-x}}{2}. \quad (1.1.2)$$

Find the derivatives $\frac{d}{dx}(\cosh x)$ and $\frac{d}{dx}(\sinh x)$ by differentiating the two definitions.

workspace

1.2 Graph, Domain, and Range of $\cosh x$ and $\sinh x$

Since $\sinh x$ and $\cosh x$ are built from the two exponentials $\frac{1}{2}e^x$ and $\frac{1}{2}e^{-x}$, their graphs are easiest to understand by sketching those exponentials first. See [Figure 1.2.1](#) and [Figure 1.2.2](#).

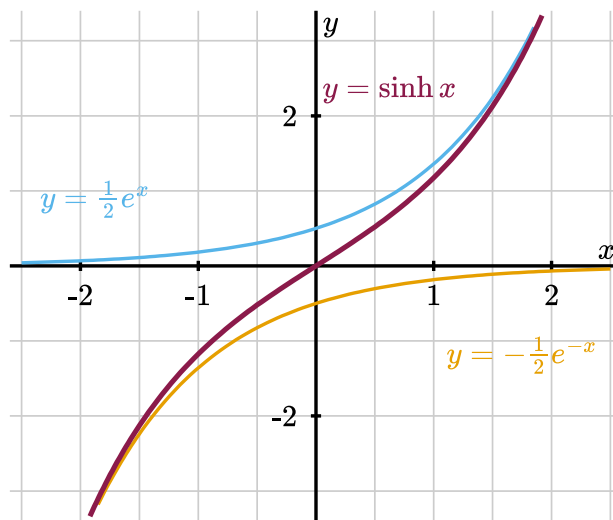


Figure 1.2.1 The graph of $y = \sinh x$. The curve is squeezed between the two exponentials $y = \frac{1}{2}e^x$ and $y = -\frac{1}{2}e^{-x}$, approaching the first as $x \rightarrow \infty$ and the second as $x \rightarrow -\infty$.

From the graph, read off the domain, the range, and the symmetry of $\sinh x$.

Domain:

Range:

Symmetry:

The graph suggests that $\sinh x$ hugs $\frac{1}{2}e^x$ when x is large and positive, and hugs $-\frac{1}{2}e^{-x}$ when x is large and negative. Confirm this observation.

workspace

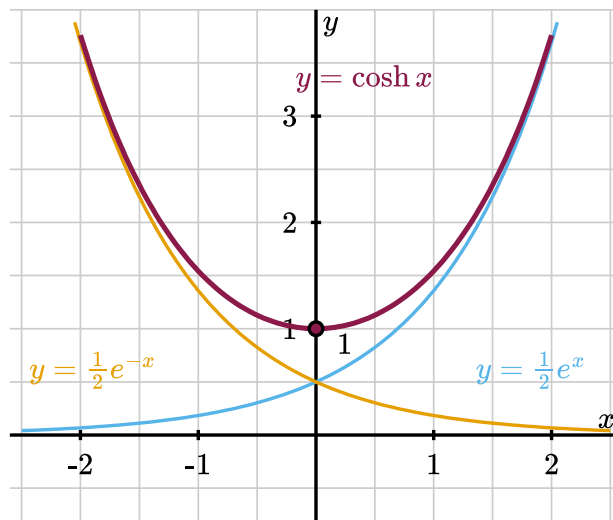


Figure 1.2.2 The graph of $y = \cosh x$. The curve is the sum of the two exponentials $y = \frac{1}{2}e^x$ and $y = \frac{1}{2}e^{-x}$, so it lies above both and has its minimum value 1 at $x = 0$.

From the graph, read off the domain, the range, and the symmetry of $\cosh x$.

Domain:

Range:

Symmetry:

The graph suggests that $\cosh x$ hugs $\frac{1}{2}e^x$ when x is large and positive, and hugs $\frac{1}{2}e^{-x}$ when x is large and negative, while never dropping below 1. Confirm these observations.

workspace

1.3 Identities and Other Hyperbolic Functions

A similar identity to the trigonometric identity $\cos^2 x + \sin^2 x = 1$ holds for the hyperbolic functions:

$$\cosh^2 x - \sinh^2 x = 1. \quad (1.3.1)$$

As you already know, any point $P(x, y)$ on the circumference of the unit circle ($x^2 + y^2 = 1$) can be described in terms of sine and cosine of an angle $t \in [0, 2\pi)$, i.e. $x = \cos t$ and $y = \sin t$, which results in the trigonometric identity $\cos^2 t + \sin^2 t = 1$. Mark this on the circle in [Figure 1.3.1](#).

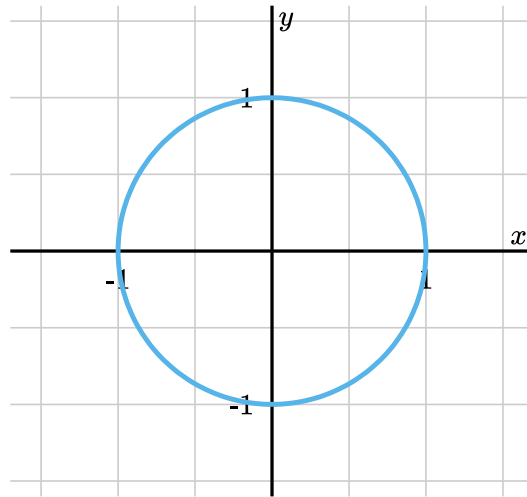


Figure 1.3.1 The unit circle $x^2 + y^2 = 1$. Mark a point $P(\cos t, \sin t)$ on it, and draw the right triangle with legs x and y and hypotenuse 1.

Similarly, any point $P(x, y)$ on the right branch of the hyperbola $x^2 - y^2 = 1$ can be represented as $x = \cosh t$ and $y = \sinh t$, where $t \in \mathbb{R}$. This follows directly from identity (1.3.1), i.e. $\cosh^2 t - \sinh^2 t = 1$. Mark this on the hyperbola in [Figure 1.3.2](#).

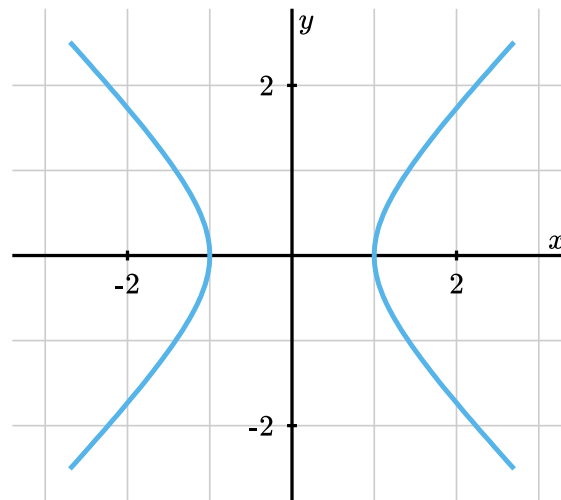


Figure 1.3.2 The hyperbola $x^2 - y^2 = 1$. Mark a point $P(\cosh t, \sinh t)$ on the right branch, and draw the segments from the origin to P and to the vertex.

Checkpoint 1.3.3 Parametrizing the Left Branch. Both branches of $x^2 - y^2 = 1$ appear in Figure 1.3.2, but the parametrization $P(\cosh t, \sinh t)$ traces only the right one, since $\cosh t \geq 1$ for every $t \in \mathbb{R}$. How would you parametrize the *left* branch, where $x \leq -1$?

Your answer and reasoning.

workspace

Hint. The identity $\cosh^2 t - \sinh^2 t = 1$ is unaffected if you change the sign of the first coordinate.

As you might have guessed, the rest of the hyperbolic functions are defined as follows.

$$\begin{aligned} \tanh x &= \frac{\sinh x}{\cosh x} & \operatorname{sech} x &= \frac{1}{\cosh x} \\ \coth x &= \frac{\cosh x}{\sinh x} & \operatorname{csch} x &= \frac{1}{\sinh x} \end{aligned}$$

1.4 Hyperbolic Identities

Below are some identities that you may find useful in some problems, however, you are not expected to memorize them. We will prove some of them as an

exercise later.

$$\cosh^2 x - \sinh^2 x = 1 \quad (1.4.1)$$

$$1 - \tanh^2 x = \operatorname{sech}^2 x \quad (1.4.2)$$

$$\sinh 2x = 2 \sinh x \cosh x \quad (1.4.3)$$

$$\cosh 2x = \cosh^2 x + \sinh^2 x \quad (1.4.4)$$

$$\cosh^2 x = \frac{\cosh 2x + 1}{2} \quad (1.4.5)$$

$$\sinh^2 x = \frac{\cosh 2x - 1}{2} \quad (1.4.6)$$

Example 1.4.1 Computing $\cosh^{-1}(x)$. Show that the inverse hyperbolic cosine can be written in terms of the natural logarithm as

$$\cosh^{-1}(x) = \ln \left(x + \sqrt{x^2 - 1} \right), \quad x \geq 1.$$

Solution. (Restrict $\cosh$ to $x \geq 0$ so that it is one-to-one, set $y = \cosh(x)$, and solve for x .)

workspace



Example 1.4.2 An integral via a hyperbolic substitution. Use hyperbolic functions to calculate the integral

$$\int \frac{\sqrt{x^2 - 1}}{x^2} dx, \quad x \geq 1.$$

Hint: Similar to $\int \sec^2(x) dx = \tan(x) + C$, we have $\int \operatorname{sech}^2(x) dx = \tanh(x) + C$.

Solution.

workspace



1.5 Hyperbolic Functions in Real Life

Hyperbolic functions are more than a formal analogue of the trigonometric functions: they describe a number of concrete physical situations. We look at three of them.

1.5.1 Hanging Cables and the Catenary

If a heavy flexible cable (such as a telephone line, a power line, or a chain) hangs freely from two supports, it settles into a curve called a **catenary**. Contrary to a common guess, this curve is not a parabola; it is the graph of a hyperbolic cosine,

$$y = a \cosh\left(\frac{x}{a}\right),$$

where x is measured horizontally from the lowest point of the cable, so that the y -axis is the axis of symmetry and the lowest point sits at height $y(0) = a$. Sliding the curve up or down, as in $y = c + a \cosh(x/a)$, only changes where we draw the x -axis.

The shape comes out of a balance of forces. Look at the piece of cable running from the lowest point to a point where the cable makes an angle θ

with the horizontal, and let s be the arc length of that piece. Three forces act on it: the tension T_0 at the lowest point, which is horizontal; the tension T along the cable at the other end; and the weight $\lambda g s$ of the piece, where λ is the mass per unit length; see Figure 1.5.1. Balancing the horizontal and the vertical components gives

$$\begin{aligned} T \cos \theta &= T_0 \\ T \sin \theta &= \lambda g s. \end{aligned}$$

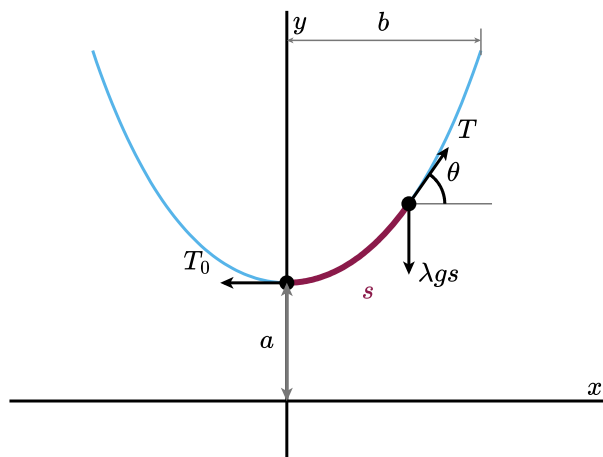


Figure 1.5.1 The three forces on the piece of cable of arc length s running from the lowest point of the cable to a point where the cable makes an angle θ with the horizontal. The lowest point sits at height $a = T_0/(\lambda g)$, and the supports are a distance $2b$ apart. (After Fig. 1 of Behroozi, cited below.)

The quantity $a = T_0/(\lambda g)$ has units of length, and dividing the second equation by the first eliminates T and leaves $s = a \tan \theta$. Since the cable makes the angle θ with the horizontal, $\tan \theta$ is its slope, so

$$\frac{dy}{dx} = \tan \theta = \frac{s}{a}.$$

Now use the arc length element $ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{1 + (s/a)^2} dx$ and separate the variables:

$$dx = \frac{a ds}{\sqrt{a^2 + s^2}} \quad \Longrightarrow \quad x = a \sinh^{-1}\left(\frac{s}{a}\right),$$

where the constant of integration vanishes because $s = 0$ at $x = 0$.

Solving for s gives $s = a \sinh(x/a)$, and therefore

$$\frac{dy}{dx} = \frac{s}{a} = \sinh\left(\frac{x}{a}\right).$$

One last integration, together with $y(0) = a$, produces the catenary equation $y = a \cosh(x/a)$.

Two things are worth noticing. First, a is a pure scale factor: written as $y/a = \cosh(x/a)$, the equation shows that every catenary is a scaled copy of the single curve $y = \cosh x$, in exactly the same way that every circle is a scaled copy of the unit circle; Figure 1.5.2 shows four of them. Second, a is fixed by the cable itself. If the cable has half-length l and its two supports are a distance $2b$ apart, then putting $s = l$ at $x = b$ in $s = a \sinh(x/a)$ gives

$$l = a \sinh\left(\frac{b}{a}\right),$$

which determines a (numerically) from the two lengths. Since $a = T_0/(\lambda g)$, a cable pulled tight has a large a and hangs almost flat, while a slack one has a small a and sags sharply.

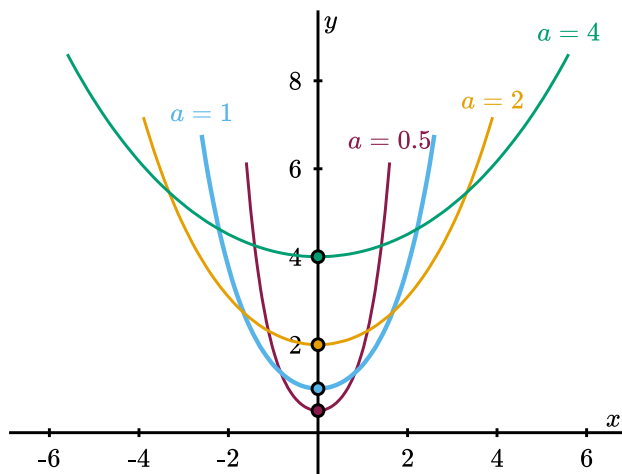


Figure 1.5.2 The catenaries $y = a \cosh(x/a)$ for $a = 0.5, 1, 2, 4$. Each curve meets the y -axis at its own value of a , and all four are scaled copies of the single curve $y = \cosh x$. (After Fig. 2 of Behroozi, cited below.)

A worked example of a hanging cable, in which we find the slope of the cable and the angle at which it meets its pole, appears in [Example 1.6.1](#).

The derivation above follows F. Behroozi, [In Praise of the Catenary](#), *The Physics Teacher* **56**, 214–217 (2018), which also discusses the sense in which all catenaries are similar to one another and suggests simple classroom demonstrations.

1.5.2 Celestial Mechanics

If a comet has enough speed, it can escape the gravitational pull of the sun, in which case one possible trajectory is a hyperbolic trajectory. The comet [2I/Borisov](#), discovered in 2019, is the first comet known to have come from outside our solar system. It was moving too fast for the sun to capture it, so its path is a hyperbola rather than an ellipse: it swung around the sun once and is now on its way back out. See [Figure 1.5.3](#).

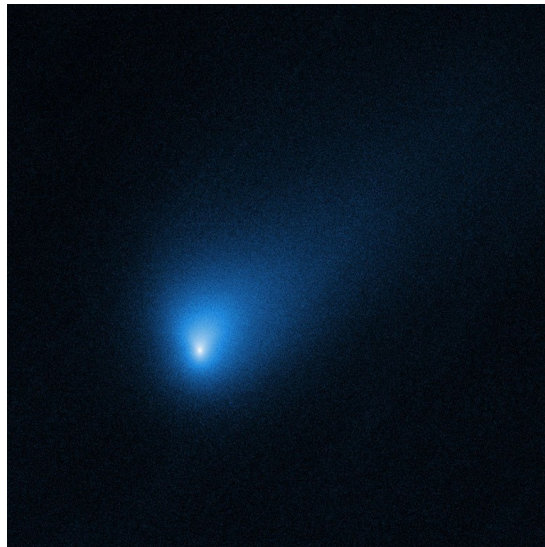


Figure 1.5.3 The interstellar comet 2I/Borisov, photographed by the Hubble Space Telescope in 2019. Because its speed exceeds the escape speed of the sun, its trajectory is a hyperbola and it passes through the solar system only once. (Image: NASA, ESA and D. Jewitt (UCLA).)

1.5.3 The Gateway Arch

The Gateway Arch in St. Louis, Missouri (designed in 1963 and completed in 1965) is a catenary turned upside down: flipping the curve converts the tension carried by a hanging chain into pure compression, which is what masonry and steel carry best. The geometric form of the gateway was set by Hannskari Bandel (structural engineer) and was expressed in the blueprints by the equation

$$y = A \left(\cosh \frac{Cx}{L} - 1 \right),$$

where A , C , and L are constants. The arch is slightly flattened compared with a uniform hanging chain, because it is thicker at the base than at the top. It stands 630 feet tall and 630 feet wide at the base; the National Park Service describes its construction and its geometry at [Gateway Arch National Park](#). See [Figure 1.5.4](#).



Figure 1.5.4 The Gateway Arch in St. Louis, Missouri. Its centerline follows the curve $y = A(\cosh(Cx/L) - 1)$, an upside-down catenary. (Photograph by John Margolies, 1988; John Margolies Roadside America photograph archive, Library of Congress, Prints and Photographs Division.)

1.6 Further Examples

Example 1.6.1 A hanging telephone line. A telephone line hangs between two poles 14 m apart in the shape of the catenary $y = 20 \cosh(x/20) - 15$, where x and y are measured in meters.

- A. Find the slope of this curve where it meets the right pole.
- B. Find the angle θ between the line and the pole.

Solution. (A sketch of the line between the poles at $x = -7$ and $x = 7$ may help.)

workspace

Example 1.6.2 Rewriting $\sinh(\ln(x))$. Consider the function $f(x) = \sinh(\ln(x))$. ▲

A. Express $f(x)$ as a fraction of two polynomials.

B. Calculate $\frac{df}{dx}$.

Solution.

workspace



Example 1.6.3 A double-angle identity. Prove that $\sinh(2x) = 2 \sinh(x) \cosh(x)$.

Proof (start from the definitions).

workspace



Example 1.6.4 Solving a hyperbolic equation. Consider the equation $2 \cosh(2x) + 10 \sinh(2x) = 5$ and solve for x .

Solution.

workspace



Chapter 2

Assignments and Review Problems

The ten assignments and the four review problem sets, in one place. Worked solutions appear beneath each problem once the assignment's due date has passed; solutions for the review sets are posted the same way.

Using the AI Tutor.

Stuck? A Socratic AI tutor is available on every assignment page. It knows the problems on the page and will coach you with questions and gentle hints — but it will never give away an answer. Here is how to use it.

Opening the tutor. On an assignment page, click the round “Tutor” button in the lower-right corner. A chat panel opens and stays in place as you scroll, so you can read a problem and talk to the tutor at the same time. The dash button in the panel's title bar minimizes it back to the button.

The tutor is available on each of the ten assignment pages. The review problem sets do not have it: those sets are practice for the exams, where you will be entirely on your own.

One-time setup. The first time you open the tutor it asks for a free Google AI Studio key (this takes about a minute):

1. Go to aistudio.google.com/apikey and sign in with a *personal* Google account — a regular Gmail. Do *not* use your @scu.edu account: the university has this service turned off, and you will see “Your account is managed by an organization.”
2. Click *Create API key* and copy it.
3. Paste it into the tutor.

The key is stored only in your own browser and is sent only to Google; this site never sees it. Treat the key as yours alone: anyone who has it can spend your free quota, so do not share it or paste it anywhere else.

Privacy: what to keep out of the chat. Everything you type into the tutor is sent to Google and handled under Google's terms, not the university's. On the free tier your conversations *may be read by people at Google* and used to improve their products, and they are not

covered by the protections your school records have. Nothing you type is private, so treat the panel as a public place and keep it to mathematics.

- Do not type anything that identifies you: your name, student ID, SCU email, phone number, address, or date of birth. The tutor does not need to know who you are in order to help with a gradient.
- Do not type anything about anyone else — a classmate’s name, their work, or their grade. That is their information to share, not yours.
- Keep out anything sensitive or unrelated to the course: health, finances, immigration or visa status, family matters, passwords, or anything you would not want quoted back to you.
- Before you paste or upload, look at what you are sending. A screenshot or a photo of your work can carry your name, a page header, or a browser tab in the background.
- The tutor is a study aid, not a counselor and not the instructor. If something personal is affecting your work in this course, bring it to office hours or to the university’s student services instead — those conversations belong with a person, and they stay confidential.

None of this is a reason to avoid the tutor. Ask it anything about the mathematics — what you tried, where you are stuck, why a step feels wrong. Just keep yourself, and everyone else, out of the transcript.

Working with the tutor. Choose the problem you are working on from the drop-down at the top of the panel, then tell the tutor what you have tried so far — the more of your own thinking you show, the more useful its questions become. The tutor can nudge you toward a starting point, and it can point you to sections and worked examples in these notes to study.

If your answer is wrong, the tutor will tell you so and help you understand *why* it is wrong: which step breaks, what misconception is behind it, and what picture or small example will let you see the failure for yourself. Note that this requires showing your reasoning — a bare answer with no work, right or wrong, will simply get asked for the thinking behind it. And what the tutor will not do is tell you how to correct an error — no fixed-up step, no “instead, you should...” Finding the right path once you understand why the old one fails is your part of the work.

Two models. Next to the problem picker is a “Model” drop-down offering *Careful* and *Quick*. Both are Gemini Flash models. Careful is the default: it thinks before it answers, so it takes a few seconds longer and is better at staying in the role of a tutor rather than a solutions manual. Quick is lighter and replies almost instantly. Either will coach you and neither will hand you an answer; switching between them costs nothing and does not disturb your conversation or your hint count. Try both — and tell me which you prefer.

Hints are limited on purpose. The point of the tutor is to get you unstuck, not to walk you to the answer, so hints are deliberately scarce. The exact rules:

- You get at most *three hints per problem, total* — the budget never resets. The panel always shows how many you have left on the problem you selected (“Hints left: 2/3”).
- *Every* reply from the tutor costs a hint — there are no free ones. Lead with what you have tried, so each reply can meet your actual thinking instead of spending itself asking for it.
- Restarting the conversation, reloading the page, or switching problems does not refill hints. Spend them thoughtfully — full solutions are posted after the due date.
- When your hints for a problem run out, bring your work to office hours — working through it together is exactly what office hours are for.

Tell me how it is going. Under every hint is a single question — “Was this hint helpful?” — with a thumb up and a thumb down. One tap is the whole thing, and it is the most useful second you can spend on this course: it tells me which hints land and which ones miss, hint by hint. Nothing but your answer and which problem you were on is sent.

For anything longer there is the “feedback” button at the top of the panel, which sends me a short note about how the tutor is working for you: whether it helped, and — in a box you can write as much as you like in — what would make it better. Tapping the thumb down offers it too. All of it is anonymous, it has nothing to do with your grade, and the tutor itself never sees it. If you want me to see the exchange you just had, there is a box to attach it; it is off unless you tick it. The tutor is new, and what you write is how it gets better.

Finally, know what the tutor will and will not do. It will not state answers, complete a solution, or turn a wrong answer into a right one — no matter how you ask. There is exactly one way to get an answer confirmed: state your conclusion *together with your own justification*. If your reasoning is sound and complete, the tutor will tell you that you have it; if there is a gap, it will point at the gap and let you close it. A bare answer, right or wrong, earns nothing.

2.1 Assignment 1

These problems exercise the hyperbolic identities, the derivatives of the hyperbolic functions and the inverse hyperbolic functions. If you would like to review the material first, see [Hyperbolic Functions](#).

1. Show that $\tanh^2(x) + \operatorname{sech}^2(x) = 1$, for all real numbers x .
2. Compute $\frac{d}{dx} 3 \sinh\left(\frac{3}{x}\right)$.
3. Simplify $\cosh(\ln 2)$.
4. Solve the equation $5 \sinh(x) - 3 \cosh(x) = 3$ for x .
5. Following the method of [Example 1.4.1](#), show that the inverse hyperbolic tangent is given by

$$\tanh^{-1}(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right), \quad -1 < x < 1.$$

6. Use the substitution $x = \tanh(t)$ to calculate

$$\int \frac{dx}{1-x^2}, \quad -1 < x < 1,$$

and then use [Worksheet Exercise 2.1.5](#) to write your answer with a logarithm. *Hint:* $\frac{d}{dt} \tanh(t) = \operatorname{sech}^2(t)$, and $1 - \tanh^2(t) = \operatorname{sech}^2(t)$.